RAJALAKSHMI ENGINEERING COLLEGE

RAJALAKSHMI NAGAR, THANDALAM – 602 105



CS23331 - DESIGN AND ANALYSIS OF ALGORITHMS

Laboratory Record Note Book

Name : Raqib Ahmed MR……. ……...

Year / Branch / Section : II year Computer science and design…

University Register No. : .2116251701045……

College Roll No. : .251701045……..……

Semester : …III……..……

Academic Year : .2026…….

RAJALAKSHMI ENGINEERING COLLEGE

Raqib Ahmed M R 251701045-2025-CSD

R2

Started on	Monday, 28 September 2026, 3:23 PM
State	Finished
Completed on	Monday, 28 September 2026, 3:35 PM
Time taken	11 mins 18 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement

Given an array of 1s and 0s this has all 1s first followed by all 0s. Aim is to find the number of 0s. Write a program using Divide and Conquer to Count the number of zeroes in the given array.

Input Format

First Line Contains Integer m – Size of array

Next m lines Contains m numbers – Elements of an array

Output Format

First Line Contains Integer – Number of zeroes present in the given array.

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main() {
    int m, l = 0, h, mid;
    if (scanf("%d", &m) != 1) return 0;
    int a[m]; h = m - 1;
    for (int i = 0; i < m; i++) scanf("%d", &a[i]);
    while (l <= h) {
        mid = (l + h) / 2;
        if (a[mid]) l = mid + 1; else h = mid - 1;
    }
    printf("%d\n", m - l);
}
```

	Input	Expected	Got	
✔	5 1 1 1 0 0	2	2	✔

	Input	Expected	Got	
✔	10 1 1 1 1 1 1 1 1 1 1	0	0	✔
✔	8 0 0 0 0 0 0 0 0	8	8	✔
✔	17 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0	2	2	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course

RAJALAKSHMI ENGINEERING COLLEGE

Raqib Ahmed M R 251701045-2025-CSD

R2

Started on	Monday, 28 September 2026, 3:38 PM
State	Finished
Completed on	Monday, 28 September 2026, 3:44 PM
Time taken	6 mins 14 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array `nums` of size `n`, return *the majority element*.

The majority element is the element that appears more than `⌊n / 2⌋` times. You may assume that the majority element always exists in the array.

Example 1:

```
Input: nums = [3,2,3]
Output: 3
```

Example 2:

```
Input: nums = [2,2,1,1,1,2,2]
Output: 2
```

Constraints:

- `n == nums.length`
- `1 <= n <= 5 * 10^4`
- `-2^31 <= nums[i] <= 2^31 - 1`

For example:

Input	Result
3 3 2 3	3
7 2 2 1 1 1 2 2	2

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main() {
    int n, cand = 0, count = 0, num;
    if (scanf("%d", &n) != 1) return 0;

    for (int i = 0; i < n; i++) {
        scanf("%d", &num);
        if (count == 0) cand = num;
        count += (num == cand) ? 1 : -1;
    }

    printf("%d\n", cand);
    return 0;
}
```

	Input	Expected	Got	
✔	3 3 2 3	3	3	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course

Started on	Monday, 28 September 2026, 4:14 PM
State	Finished
Completed on	Monday, 28 September 2026, 4:15 PM
Time taken	37 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement:

Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Value for x

Output Format

First Line Contains Integer – Floor value for x

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int findFloor(int arr[], int n, int x) {
    int low = 0, high = n - 1, floor = -1;
    while (low <= high) {
        int mid = low + (high - low) / 2;
        if (arr[mid] == x) return arr[mid];
        if (arr[mid] < x) { floor = arr[mid]; low = mid + 1; }
        else high = mid - 1;
    }
    return floor;
}

int main() {
    int n, x;
    if (scanf("%d", &n) != 1) return 0;
    int arr[n];
    for (int i = 0; i < n; i++) scanf("%d", &arr[i]);
    scanf("%d", &x);
    printf("%d\n", findFloor(arr, n, x));
    return 0;
}

```

	Input	Expected	Got	
✔	6 1 2 8 10 12 19 5	2	2	✔

	Input	Expected	Got	
✔	5 10 22 85 108 129 100	85	85	✔
✔	7 3 5 7 9 11 13 15 10	9	9	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Started on	Monday, 28 September 2026, 4:16 PM
State	Finished
Completed on	Monday, 28 September 2026, 4:17 PM
Time taken	24 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement:

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as “No”.

Note: Write a Divide and Conquer Solution

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

Output Format

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value “x”)

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int findPair(int a[], int l, int h, int x) {
    if (l >= h) return 0;
    if (a[l] + a[h] == x) return printf("%d\n%d\n", a[l], a[h]), 1;
    return (a[l] + a[h] < x) ? findPair(a, l + 1, h, x) : findPair(a, l, h - 1, x);
}

int main() {
    int n, x;
    if (scanf("%d", &n) != 1) return 0;
    int a[n];
    for (int i = 0; i < n; i++) scanf("%d", &a[i]);
    scanf("%d", &x);
    if (!findPair(a, 0, n - 1, x)) printf("No\n");
    return 0;
}
```

	Input	Expected	Got	
✔	4 2 4 8 10 14	4 10	4 10	✔
✔	5 2 4 6 8 10 100	No	No	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Started on	Monday, 28 September 2026, 4:18 PM
State	Finished
Completed on	Monday, 28 September 2026, 4:18 PM
Time taken	25 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Write a Program to Implement the Quick Sort Algorithm

Input Format:
The first line contains the no of elements in the list-n
The next n lines contain the elements.

Output:
Sorted list of elements

For example:

Input	Result
5 67 34 12 98 78	12 34 67 78 98

Answer:

```
#include <stdio.h>

void q(int a[], int l, int h) {
    if (l >= h) return;
    int p = a[h], i = l - 1, t, j = l;
    for (; j < h; j++) if (a[j] < p) { t = a[++i]; a[i] = a[j]; a[j] = t; }
    t = a[++i]; a[i] = a[h]; a[h] = t;
    q(a, l, i - 1); q(a, i + 1, h);
}

int main() {
    int n, i = 0;
    if (scanf("%d", &n) > 0) {
        int a[n];
        for (; i < n; i++) scanf("%d", &a[i]);
        q(a, 0, n - 1);
        for (i = 0; i < n; i++) printf("%d ", a[i]);
    }
}
```

	Input	Expected	Got	
✔	5 67 34 12 98 78	12 34 67 78 98	12 34 67 78 98	✔
✔	10 1 56 78 90 32 56 11 10 90 114	1 10 11 32 56 56 78 90 90 114	1 10 11 32 56 56 78 90 90 114	✔
✔	12 9 8 7 6 5 4 3 2 1 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.